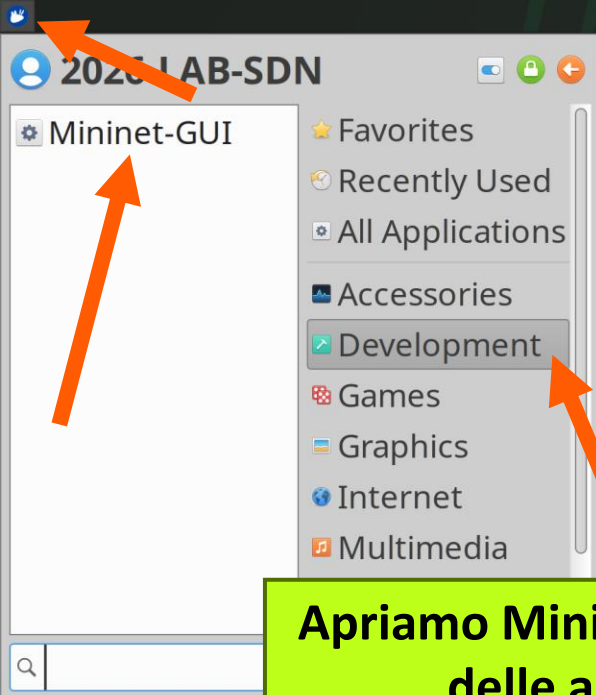


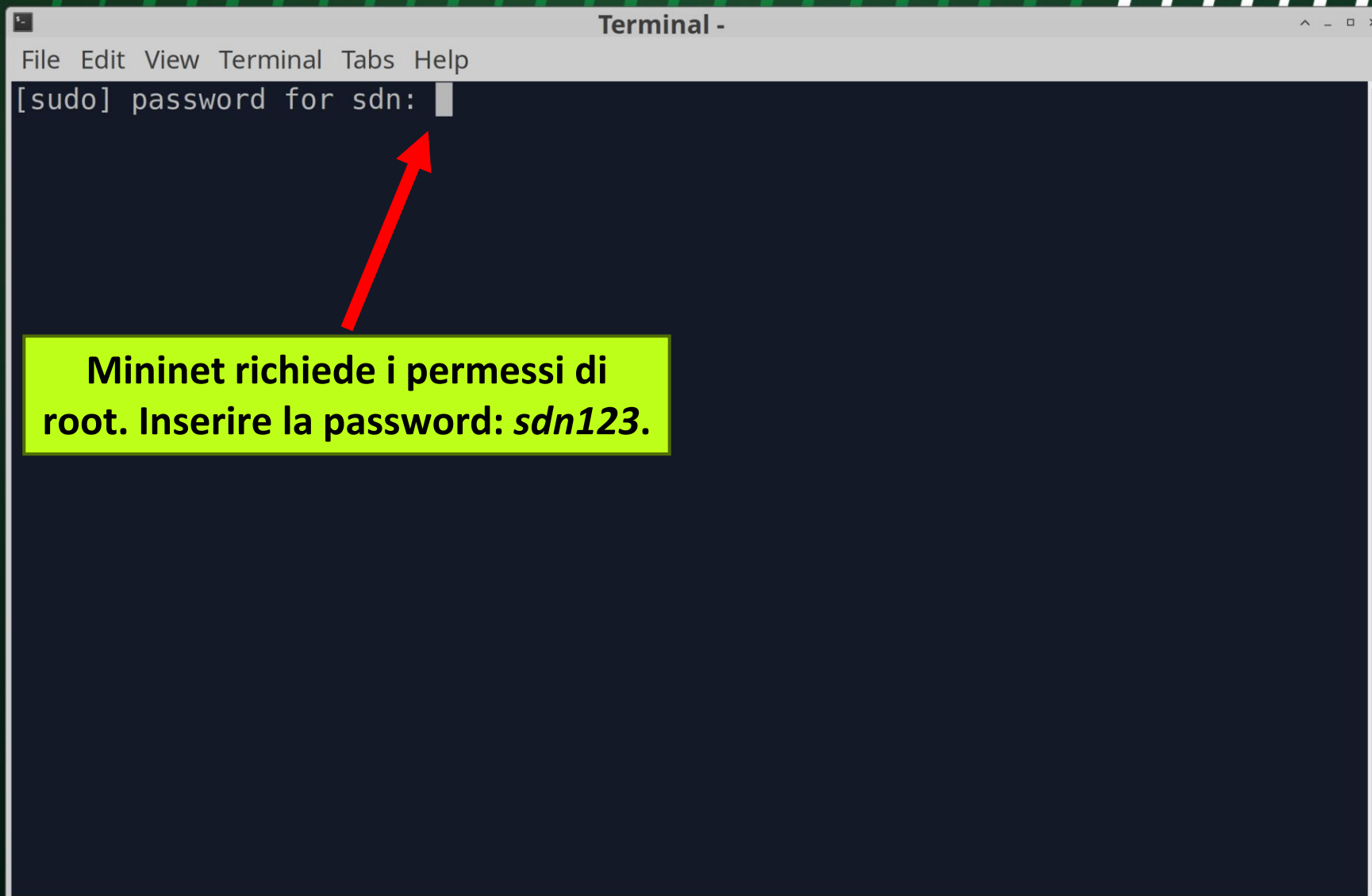
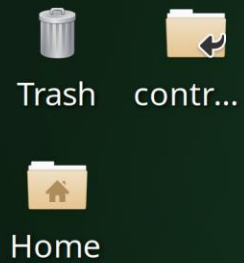


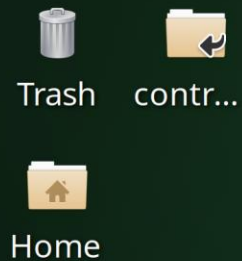
POLITECNICO
MILANO 1863

SDN Challenge #1



Apriamo Mininet-GUI dal menù delle applicazioni.





```
Terminal -
File Edit View Terminal Tabs Help
[sudo] password for sdn:
Stopping previous processes...
✓ Cleared backend log file
Starting backend...
Waiting for backend...
✓ Backend ready
Starting frontend...
Now using node v20.20.2 (npm v10.8.1)
nohup: appending output to 'nohup.out'

Mininet-GUI is available at:
  http://10.0.2.15:4020
  http://192.168.56.105:4020
Press Enter to close...
```

**Mininet-GUI è ora in ascolto.
Premiamo sul primo link
(CTRL+Click) per aprire la GUI.**

Mininet-GUI

Not Secure http://10.0.2.15:4020

90%

Mininet GUI

File

View

Nodes Palette

MAIN NODES

Host

Switch

Controller

Rem

Remote

Trasciniamo i nodi per costruire una semplice rete: 1x Switch, 2x Host, 1x Controller.

Network Tools

WebShell

Sniffer

Monitoring

Logs

Connected to Mininet

Network Stopped

Hosts 0 · Controllers 0 · Switches 0

Mininet-GUI

Not Secure http://10.0.2.15:4020

90%

Mininet GUI

File

View

≡

Run Pingall Test

Run IPerf

Topology

Generate Topology

Connect Nodes

Delete Selected

Nodes Palette

MAIN NODES

s1

c1

h2

10.0.0.2/8

Entriamo in modalità «Connect Nodes» per effettuare le permutazioni.

Network Tools

WebShell

Sniffer

Monitoring

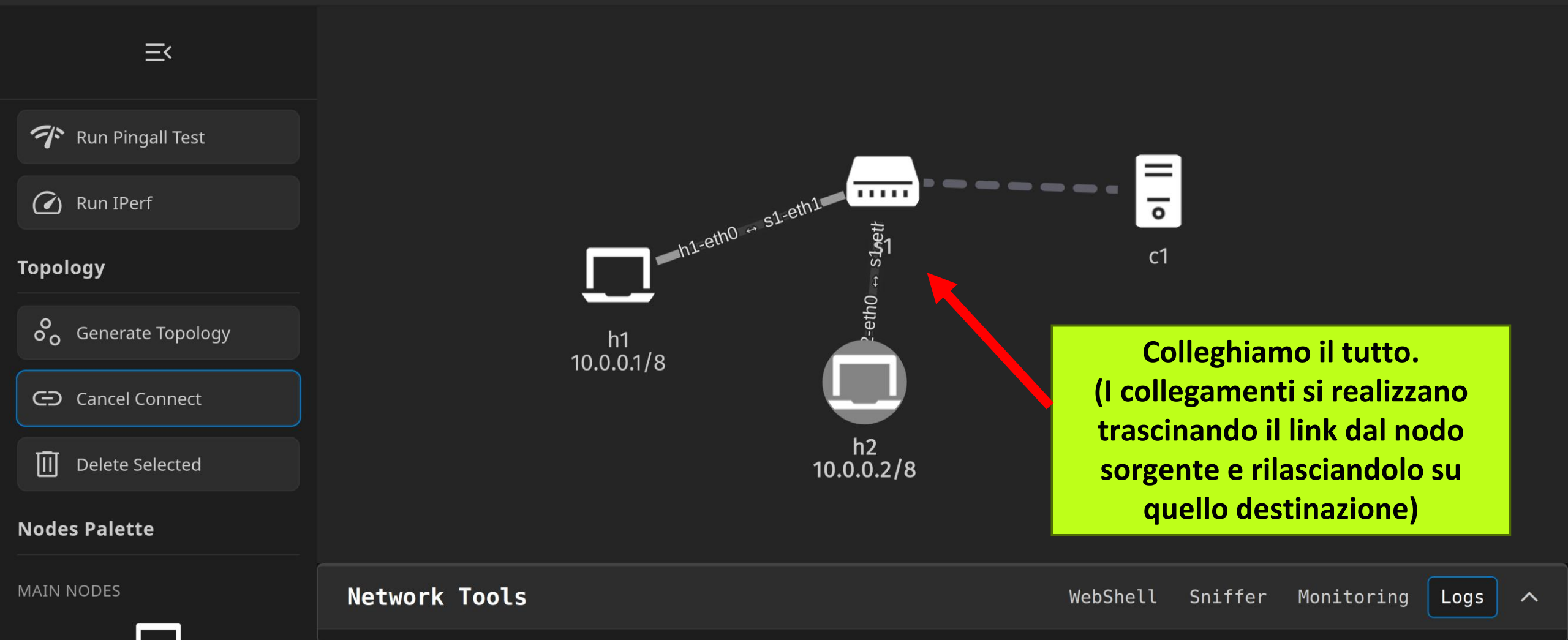
Logs

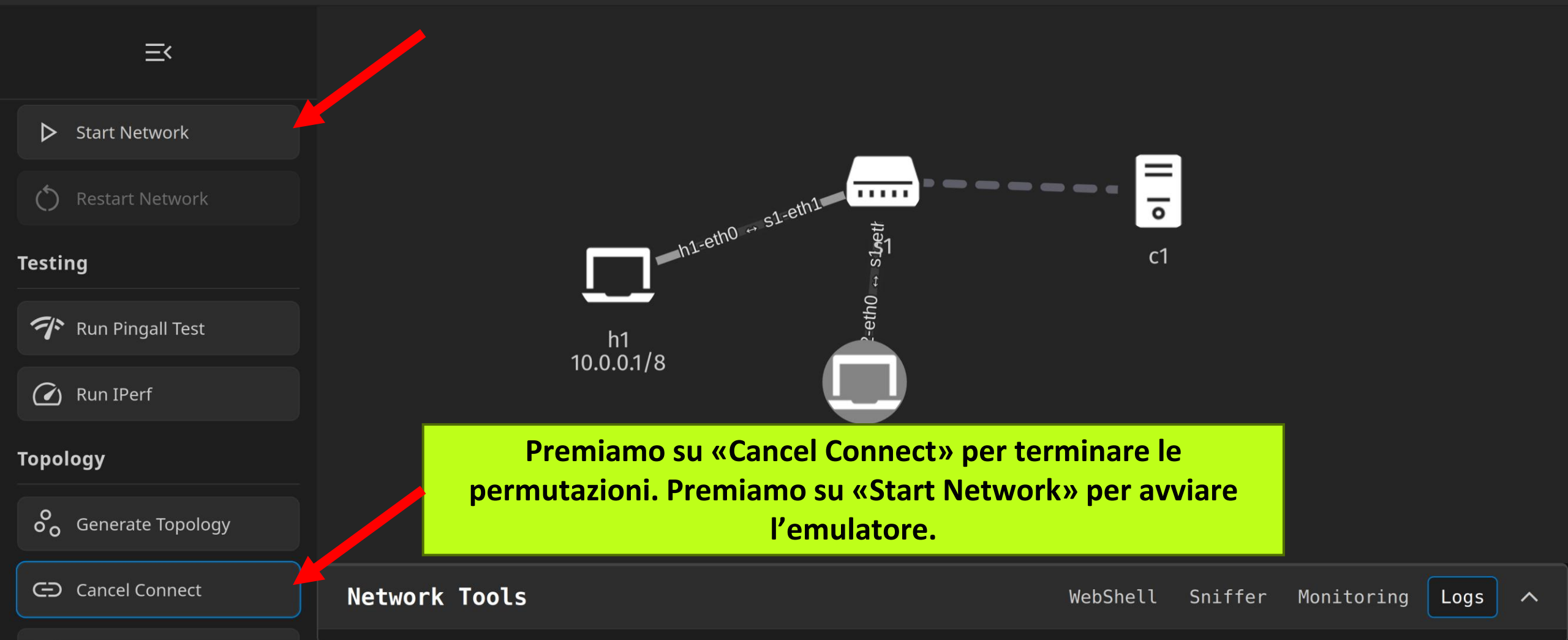
^

Connected to Mininet

Network Stopped

Hosts 2 · Controllers 1 · Switches 1







Proviamo a vedere se le macchine si parlano tra loro: «Run PingAll Test».

Run Pingall Test

Run IPerf

Generate Topology

Connect Nodes

Network Tools

WebShell

Sniffer

Monitoring

Logs

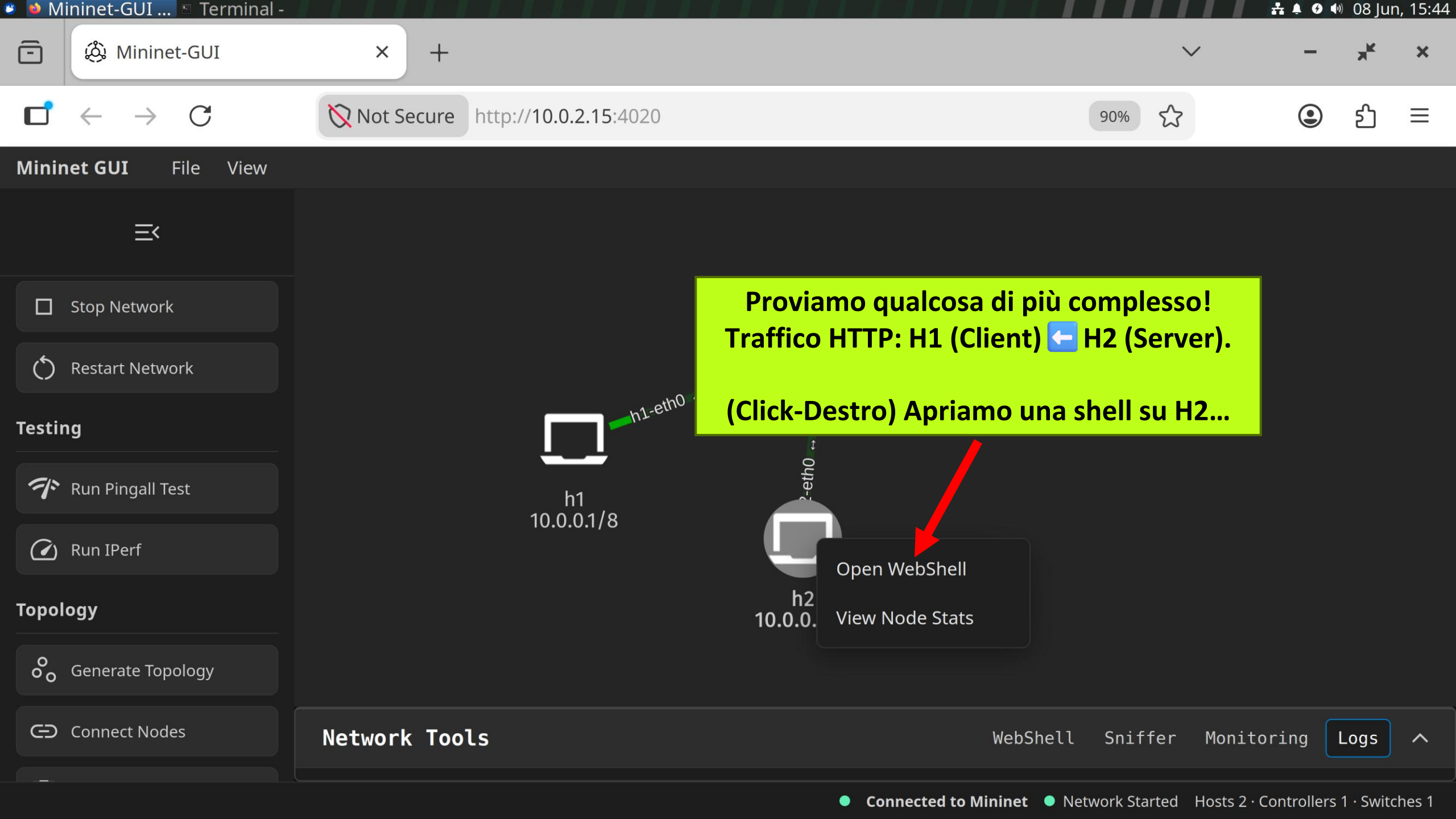
Connected to Mininet Network Started Hosts 2 · Controllers 1 · Switches 1

≡<
Stop Network
Restart Network
Testing
Run Pingall Test
Run IPerf
Topology
Generate Topology
Connect Nodes

Le macchine riescono a parlarsi!
(i.e. C'è continuità a livello IP, L3!)

Pingall Results

SOURCE	DESTINATION	SENT	RECEIVED	MIN (MS)	AVG (MS)	MAX (MS)	MDEV (MS)
h1	h2	1	1	39.436	39.436	39.436	0.000
h2	h1	1	1	20.388	20.388	20.388	0.000



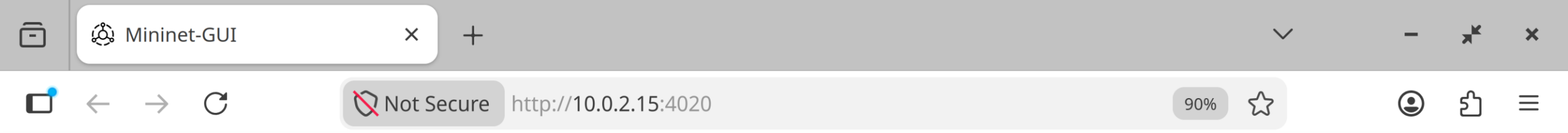
Proviamo qualcosa di più complesso!
Traffico HTTP: H1 (Client) ← H2 (Server).
(Click-Destro) Apriamo una shell su H2...

Open WebShell
View Node Stats

Network Tools

WebShell Sniffer Monitoring Logs

Connected to Mininet Network Started Hosts 2 · Controllers 1 · Switches 1



Mininet GUI File View

⋮

❑ Stop Network

🔄 Restart Network

Testing

📶 Run Pingall Test

📈 Run IPerf

Topology

🔗 Generate Topology

🔗 Connect Nodes

h1 10.0.0.1/8 s1 c1

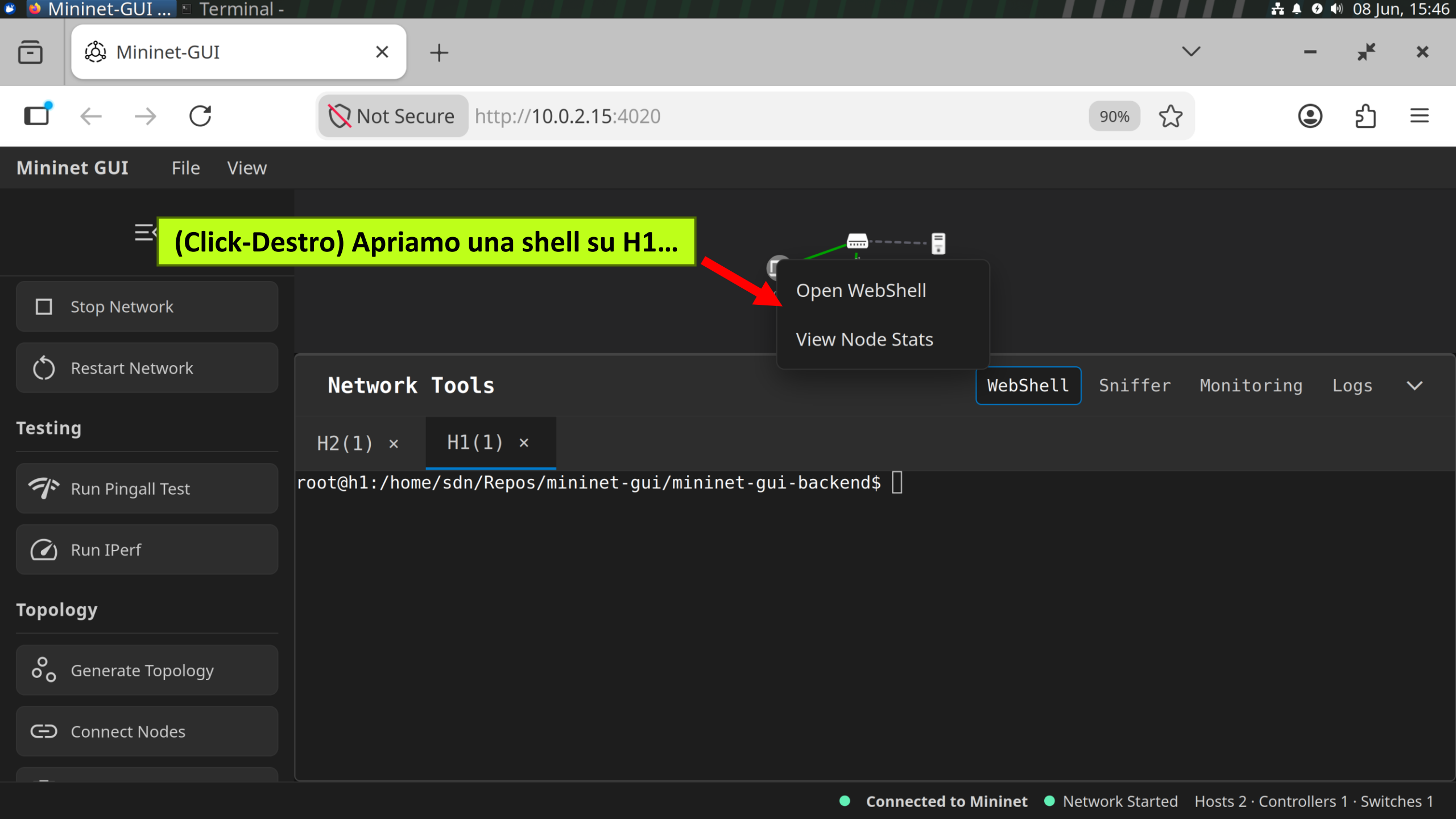
... e avviamo il server HTTP.

Network Tools WebShell Sniffer Monitoring Logs

H2(1) x

```
root@h2:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ cd ..
root@h2:/home/sdn/Repos/mininet-gui$ cd utils/webserver-ttl/
root@h2:/home/sdn/Repos/mininet-gui/utils/webserver-ttl$ ./webserver-ttl
2026/06/08 15:45:21 listening on :2000 (TTL threshold: 100)
```

**Si trova in ../utils/webserver-ttl/.
Avviato ascolta sulla porta TCP 2000.**



(Click-Destro) Apriamo una shell su H1...

Open WebShell

View Node Stats

Network Tools

WebShell

Sniffer

Monitoring

Logs

H2(1) ×

H1(1) ×

```
root@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend$
```

Mininet-GUI

Not Secure http://10.0.2.15:4020

90%

Mininet GUI

File

View

≡

Stop Network

Restart Network

Testing

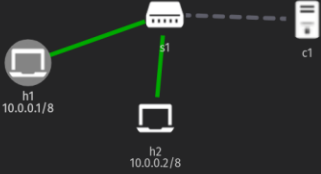
Run Pingall Test

Run IPerf

Topology

Generate Topology

Connect Nodes



```
graph LR; h1((h1  
10.0.0.1/8)) --- s1((s1)); h2((h2  
10.0.0.2/8)) --- s1; s1 -.- c1[c1];
```

Network

WebShell

Sniffer

Monitoring

Logs

H2(1) ×

H1(2) ×

root@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend\$ curl http://10.0.0.2:2000

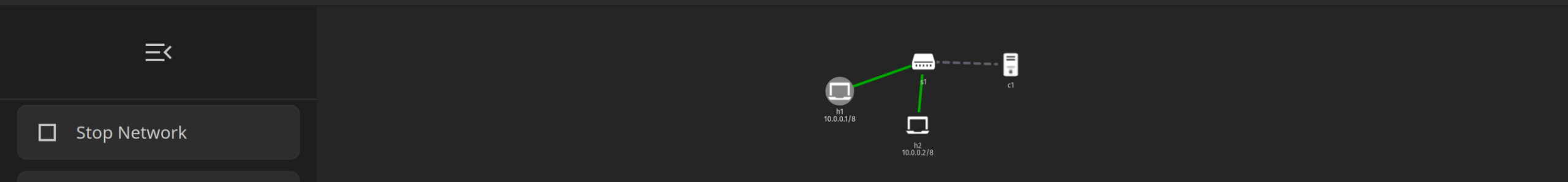
...e facciamo una richiesta HTTP a H2.

«curl» è un software che permette di fare richieste HTTP direttamente dal terminale. In questi scenari è più comodo e veloce che installare e aprire un browser.

Connected to Mininet

Network Started

Hosts 2 · Controllers 1 · Switches 1



Network Tools WebShell Sniffer Monitoring Logs

H2(1) x H1(2) x

```
root@h1: /home/sdn/Repos/mininet-gui/mininet-gui-backend$ curl http://10.0.0.2:2000
ACCESS_DENIEDroot@h1: /home/sdn/Repos/mininet-gui/mininet-gui-backend$
```

?

SDN Challenge #1



POLITECNICO
MILANO 1863



BONSAI Lab

✨ È arrivato il momento di spiegarvi la challenge! ✨

- Finora abbiamo fatto parlare due macchine connesse a uno Switch... Un compito talmente semplice che anche il Controller di default di Mininet – a cui non abbiamo detto assolutamente nulla – riesce a svolgere senza problemi.
- Perché allora il server HTTP ci ha risposto «ACCESS_DENIED»?



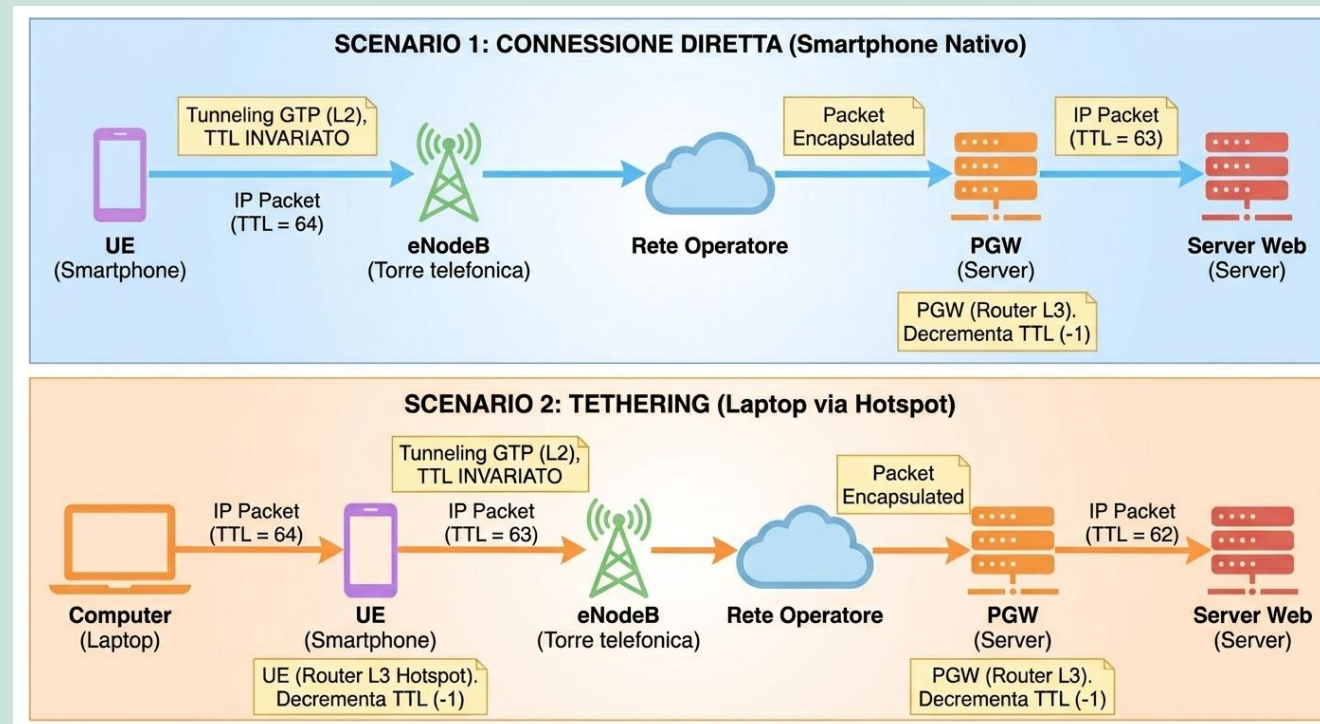
SDN Challenge #1



POLITECNICO
MILANO 1863



- Il TTL (Time to Live) è un valore nei pacchetti IP che indica il numero massimo di router (salti) che il pacchetto può attraversare prima di essere scartato, evitando che circoli all'infinito nella rete.



SDN Challenge #1

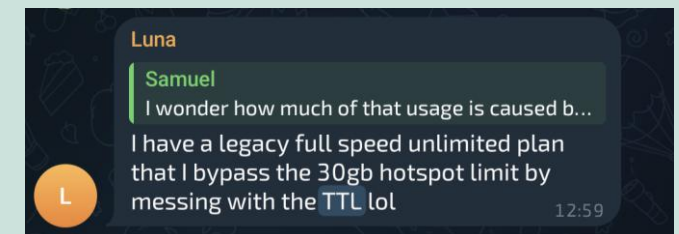
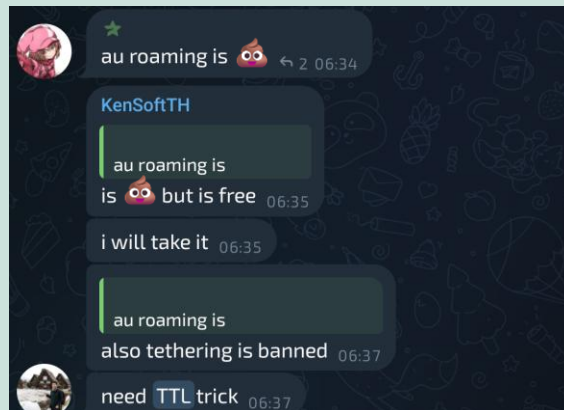


POLITECNICO
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BONSAILab

- Implicitamente questo valore ci dice quanti router un pacchetto ha attraversato...
- ... E il telefono quando fa da hotspot agisce da router...
- Sì, alcuni operatori mobili (specie in Asia) controllano fanno controllo TTL per vendere questa funzione a pagamento 💰💰.
- Sì, i clienti possono abusare della cosa e bypassare il controllo.



SDN Challenge #1



POLITECNICO
MILANO 1863



BONSAI Lab

- Il nostro webserver risponde SE E SOLO SE il TTL dei pacchetti IP in arrivo è ≥ 100 .
- 📖 Challenge: programmiamo un controller SDN in modo che istruisca gli switch sotto il suo controllo a modificare il TTL dei pacchetti a 100!
 - ⚠️ La challenge è guidata ma il punto critico è da risolvere in autonomia.
 - 🤖 Qualunque strumento è utilizzabile.



Mininet-GUI

Not Secure http://10.0.2.15:4020

90%

Mininet GUI

File

View

≡

Stop Network

Restart Network

Testing

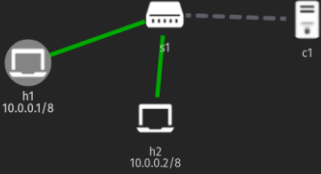
Run Pingall Test

Run IPerf

Topology

Generate Topology

Connect Nodes



```
graph TD
    h1((h1  
10.0.0.1/8)) --- s1((s1))
    h2((h2  
10.0.0.2/8)) --- s1
    s1 -.- c1[c1]
```

Network Tools

WebShell

Sniffer

Monitoring

Logs

H2(1) ×

```
root@h2:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ cd ..
root@h2:/home/sdn/Repos/mininet-gui$ cd utils/webserver-ttl/
root@h2:/home/sdn/Repos/mininet-gui/utils/webserver-ttl$ ./webserver-ttl
2026/06/08 15:45:21 listening on :2000 (TTL threshold: 100)
2026/06/08 15:46:44 rejected 10.0.0.1:55522 (TTL 64 < 100)
^C
root@h2:/home/sdn/Repos/mininet-gui/utils/webserver-ttl$
```

Connected to Mininet

Network Started

Hosts 2 · Controllers 1 · Switches 1

Mininet-GUI

Not Secure http://10.0.2.15:4020

90%

Mininet GUI

File

View

Run Pingall Test

Run IPerf

Topology

Generate Topology

Connect Nodes

Delete Selected

Nodes Palette

MAIN NODES

h1
10.0.0.1/8

h2
10.0.0.2/8

c1

s1-eth1 ↔ h1-eth0

2-eth0 ↔ s1-eth1

Network Tools

WebShell

Sniffer

Monitoring

Logs

Connected to Mininet

Network Stopped

Hosts 2 · Controllers 1 · Switches 1

Selezioniamo il «default controller» di Mininet, ed eliminiamolo «Delete Selected».

Mininet-GUI

Not Secure http://10.0.2.15:4020

90%

Mininet GUI

File

View

≡

Nodes Palette

MAIN NODES

Host

Switch

Controller

Remote

h1
10.0.0.1/8

h2
10.0.0.2/8

s1-eth1 ↔ h1-eth0

s2-eth0 ↔ s1-eth1

Aggiungiamo un controller esterno, «remoto», alla nostra rete.

Network Tools

WebShell

Sniffer

Monitoring

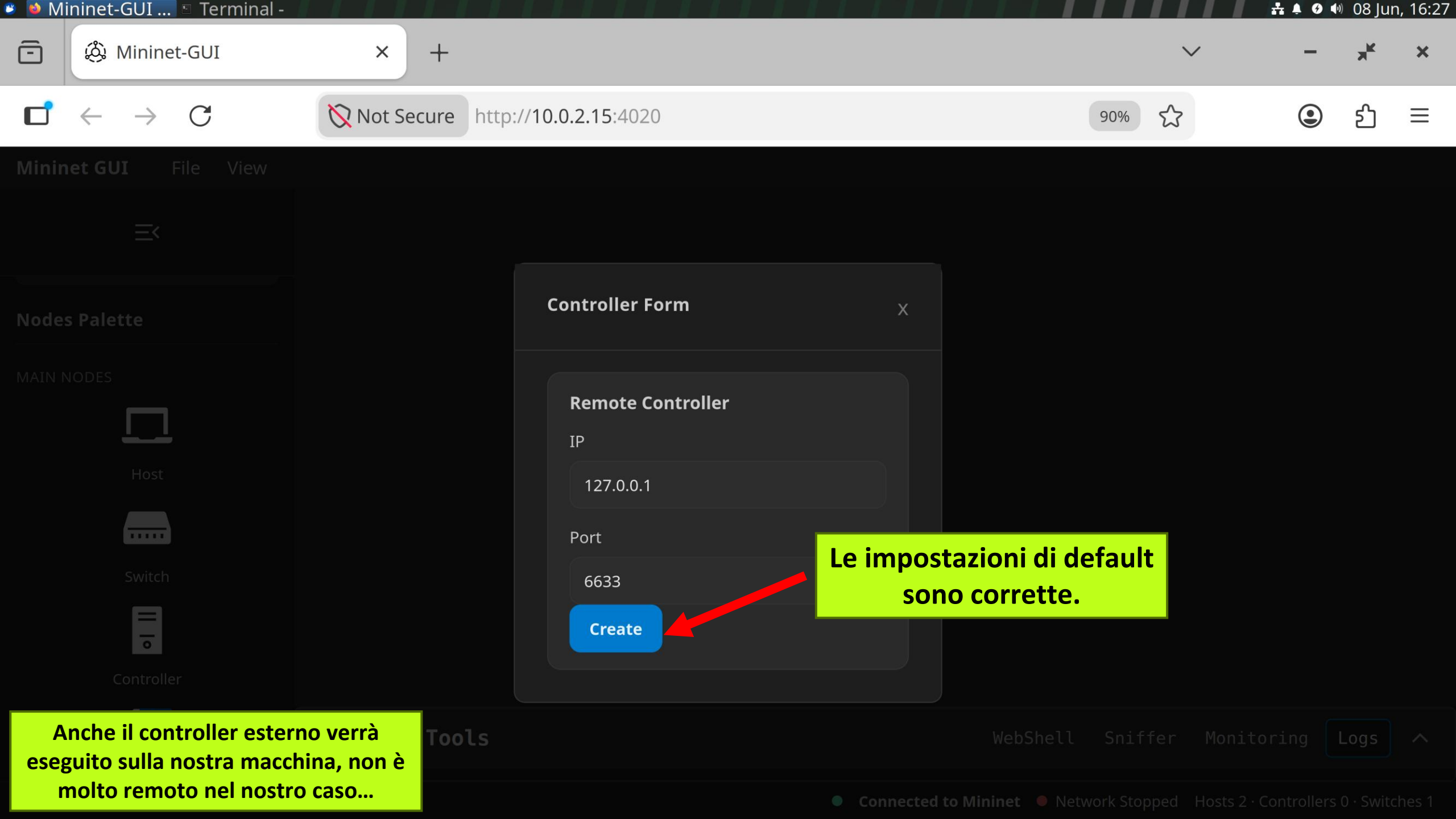
Logs

^

Connected to Mininet

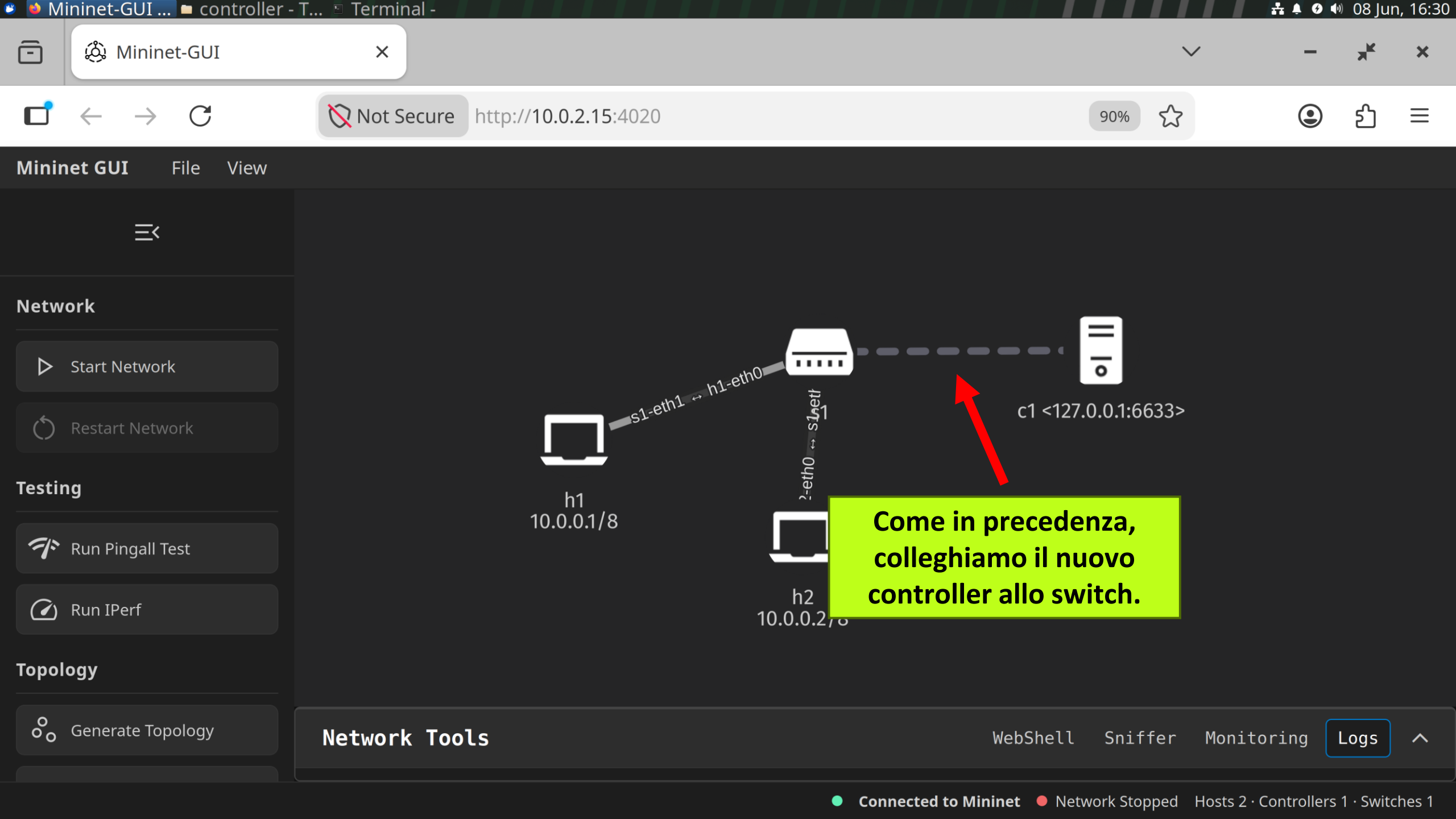
Network Stopped

Hosts 2 · Controllers 0 · Switches 1



**Le impostazioni di default
sono corrette.**

**Anche il controller esterno verrà
eseguito sulla nostra macchina, non è
molto remoto nel nostro caso...**





Trash



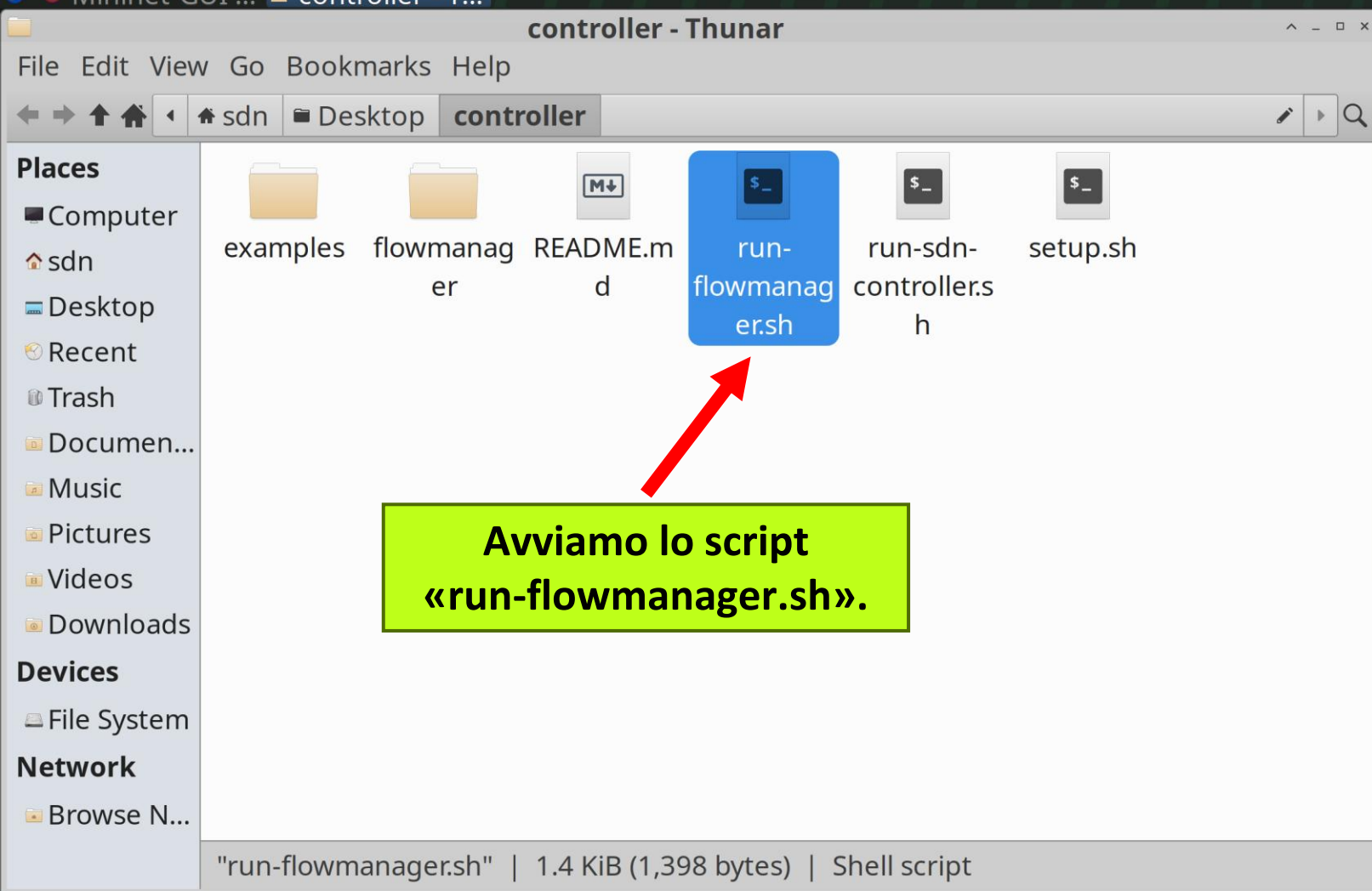
contr...

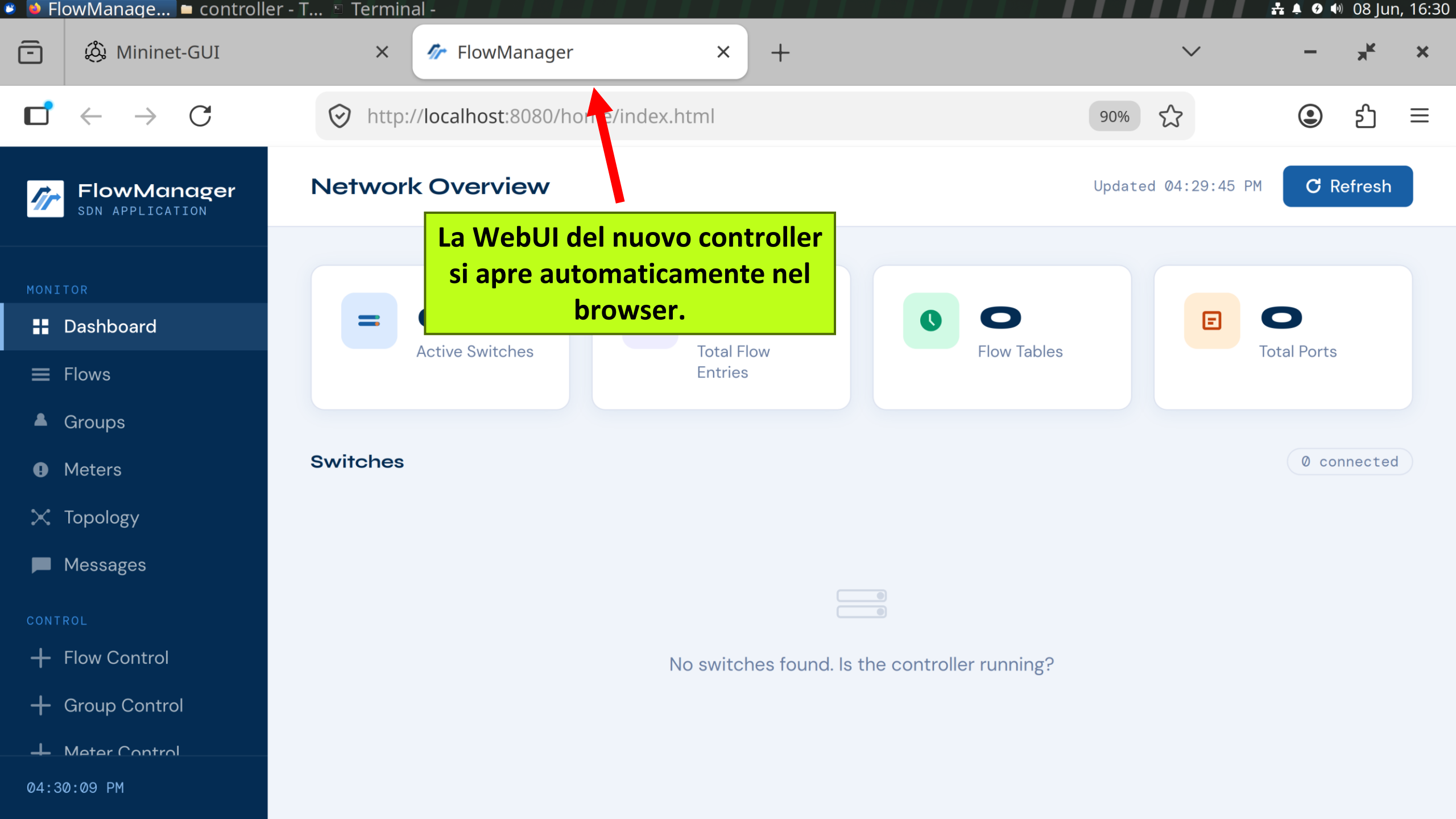


Home



Il nuovo controller, che fra non molto andremo a programmare, si trova sul Desktop nella cartella «controller».





La WebUI del nuovo controller
si apre automaticamente nel
browser.

Network Overview

Updated 04:29:45 PM

Refresh



Active Switches



Total Flow
Entries



Flow Tables



Total Ports

Switches

0 connected



No switches found. Is the controller running?

Mininet-GUI

FlowManager

Not Securehttp://10.0.2.15:402090%

Mininet GUIFileView

≡

Network

▶ Start Network

↻ Restart Network

Testing

📶 Run Pingall Test

📊 Run IPerf

Topology

🔗 Generate Topology

h1
10.0.0.1/8

h2
10.0.0.2/8

s1-eth1 ↔ h1-eth0

2-eth0 ↔ s1-eth1

c1 <127.0.0.1:6633>

Network Tools

WebShellSnifferMonitoringLogs

● Connected to Mininet

● Network Stopped

Hosts 2 · Controllers 1 · Switches 1

Riavviamo l'emulatore di rete; «Start Network».

FlowManager

Mininet-GUI

FlowManager

http://localhost:8080/home/index.html

90%

☆

👤

📌

☰

FlowManager

SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

04:30:54 PM

Network Overview

Updated 04:30:46 PM

Refresh

Active Switches

es

Total Ports

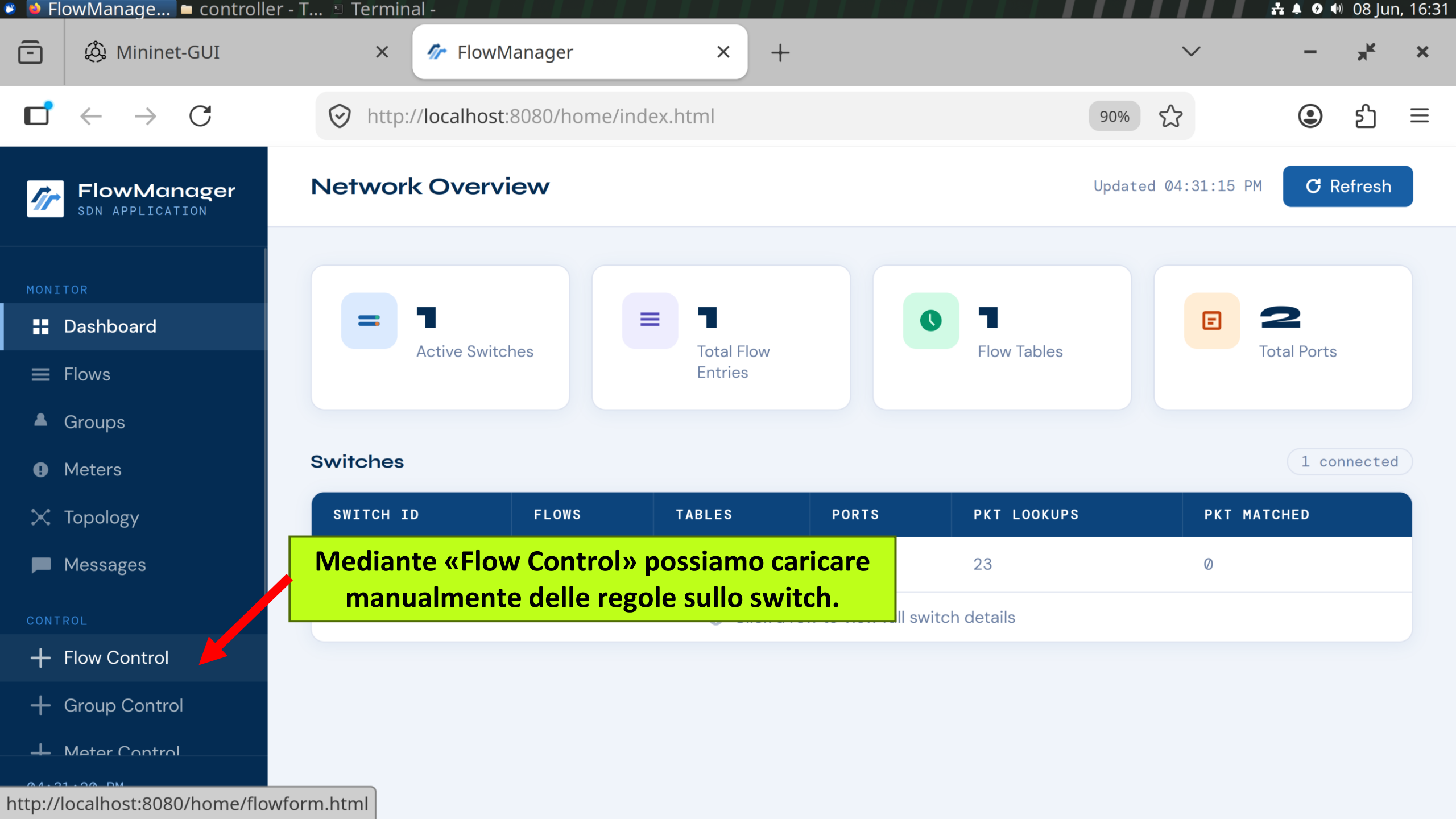
Switches

1 connected

SWITCH ID	FLOWS	TABLES	PORTS	PKT LOOKUPS	PKT MATCHED
1	1	1	2	19	0

Click a row to view full switch details

Il nuovo controller
«adotterà» il nostro switch.



SDN Challenge #1

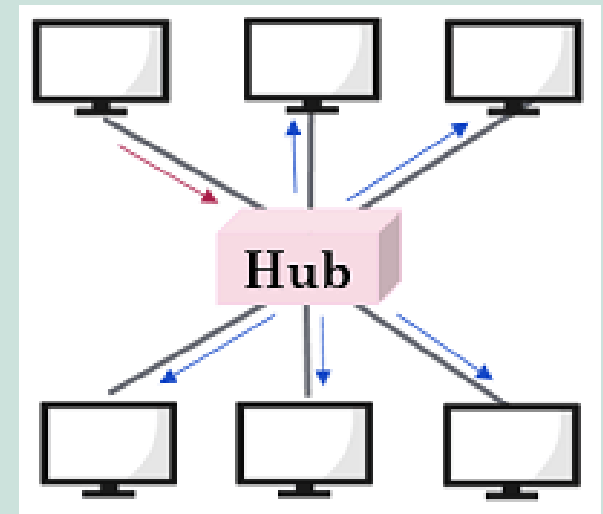


POLITECNICO
MILANO 1863



BONSAI Lab

- Osservazione: la nostra rete non ha anelli (è fatta di 1 solo switch).
- Idea: la configurazione più semplice che possiamo dare al nostro switch è quella di un HUB.
 - Regola #1 (priorità bassa): Tutto ciò che entra in una porta viene replicato su tutte le altre.
 - Regola #2 (alta priorità): Se sto processando una trama che contiene un pacchetto IP, prima di replicarla su tutte le altre porte, ne alzo il TTL a 100.
 - Abbiamo inventato un HUB che lavora a L3.



Flow Control ...

controller - T...

Terminal -

Mininet-GUI

Flow Control

http://localhost:8080/home/flowform.html

90%

FlowManager

SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

04:32:14 PM

Flow Control

1 switch available

TARGET

SwitchSW_1

Table ID

OPERATION

Add

Modify

Modify Strict

Delete

Delete Strict

PARAMETERS

Priority0

Idle Timeoutseconds

Hard Timeoutseconds

Cookieinteger

Cookie Maskinteger

FLAGS

Send flow-removed message

Check overlapping

Reset counts

Do not count packets

Do not count bytes

Regola 1: Bassa priorità...

Flow Control

1 switch available

Cookie Mask

integer

Output Port

Output Group

group id or -1 for ANY

Regola 1: ...quando entra una qualunque trama...

Q MATCH FIELDS

☒ Match Any (wildcard — ignore all match fields)

INSTRUCTIONS

Goto Meter

meter-id or -1 for All

WRITE ACTIONS

ACTION TYPE

VALUE

APPLY ACTIONS

ACTION TYPE

VALUE

e.g. OUTPUT

value

Flow Control

1 switch available

☒ Match Any (wildcard — ignore all match fields)

Regola 1: La mandiamo su tutte le porte diverse da quella d'ingresso.

Goto Met

ACTIONS

APPLY ACTIONS

ACTION TYPE

VALUE

OUTPUT

Output packet from "port"

[+ Add Action](#)

Clear Actions

FLOOD

ACTION TYPE

VALUE

e.g. OUTPUT

value

[+ Add Action](#)

Write Metadata

integer

Metadata Mask

integer

Goto Table

Flow Control ...

Pictures - Th...

Terminal -

Mininet-GUI

Flow Control

http://localhost:8080/home/flowform.html

90%

1 switch available

FlowManager

SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

04:37:56 PM

Flow Control

1 switch available

Goto Meter

meter-id or -1 for All

APPLY ACTIONS

ACTION TYPE	VALUE
OUTPUT	FLOOD
+ Add Action	
☐ Clear Actions	

WRITE ACTIONS

ACTION TYPE	VALUE
e.g. OUTPUT	value
+ Add Action	
Write Metadata	integer
Metadata Mask	integer
Goto Table	table_id

✓ Submit

↺ Clear

Regola 1: Programmiamo la regola.

Flow Control ...controller - T...Terminal -

Mininet-GUI

Flow Control

http://localhost:8080/home/flowform.html

90%

08 Jun, 16:34

FlowManager

SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

04:34:43 PM

Flow Control

1 switch available

TARGET

SwitchSW_1

Table ID

OPERATION

Add

Modify

Modify Strict

Delete

Delete Strict

PARAMETERS

Priority100

Idle Timeoutseconds

Hard Timeoutseconds

Cookieinteger

Cookie Maskinteger

FLAGS

Send flow-removed message

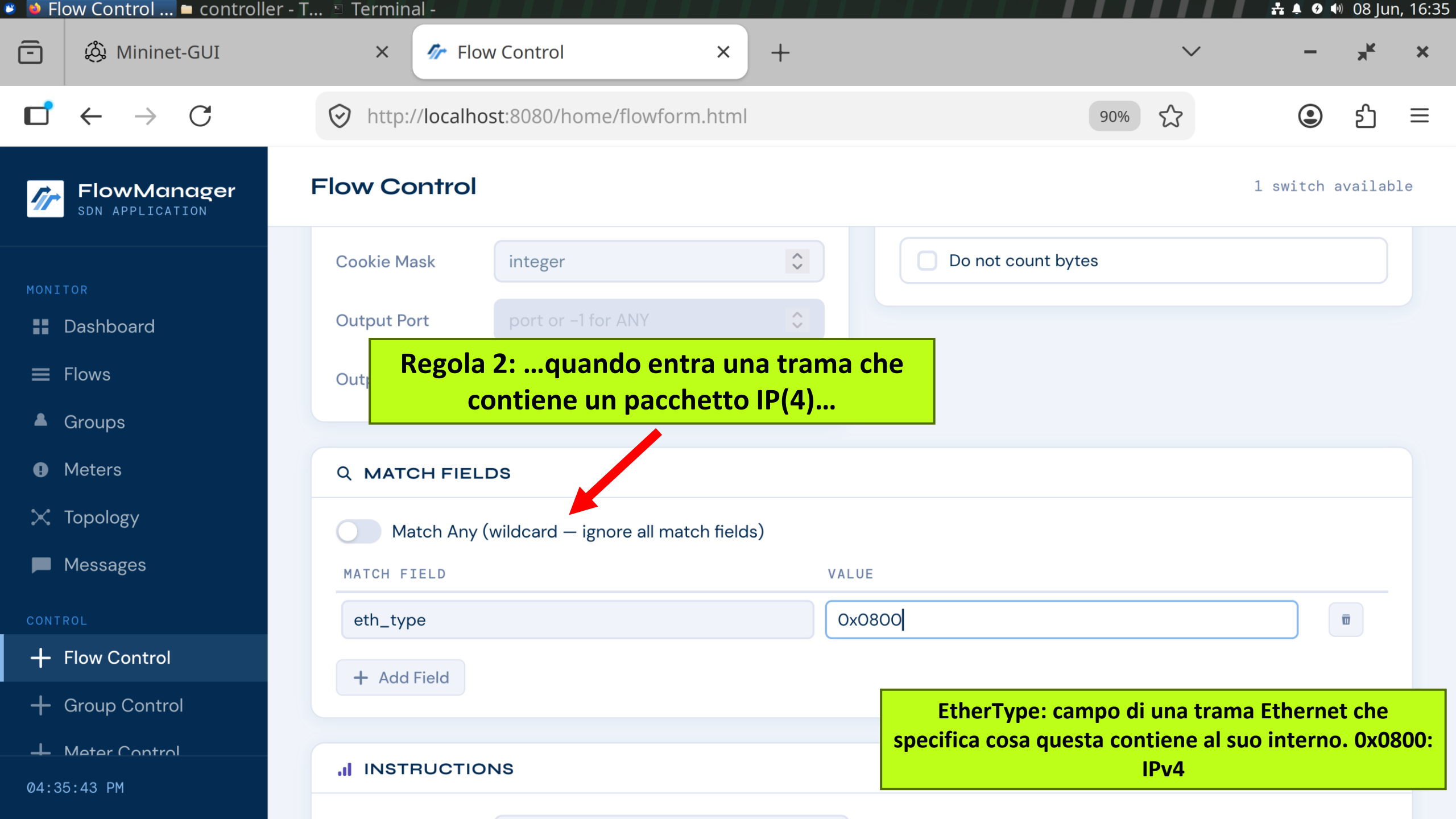
Check overlapping

Reset counts

Do not count packets

Do not count bytes

Regola 2: Alta priorità...



Flow Control

1 switch available

Cookie Mask

integer



Do not count bytes

Output Port

port or -1 for ANY

Output

Regola 2: ...quando entra una trama che contiene un pacchetto IP(4)...

MATCH FIELDS



Match Any (wildcard — ignore all match fields)

MATCH FIELD

VALUE

eth_type

0x0800

+ Add Field

EtherType: campo di una trama Ethernet che specifica cosa questa contiene al suo interno. 0x0800: IPv4

INSTRUCTIONS

Flow Control ...

Pictures - Th...

Terminal -

Mininet-GUI

Flow Control

90%

☆

👤

📌

☰

FlowManager

SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

04:37:31 PM

Flow Control

1 switch available

+ Add Field

INSTRUCT

Goto Meter

APPLY ACTIONS

ACTION TYPE	VALUE
SET_FIELD	
OUTPUT	FLOOD

+ Add Action

☐ Clear Actions

ACTION TYPE

VALUE

e.g. OUTPUT

value

🗑

+ Add Action

Write Metadata

integer

⌵

Metadata Mask

integer

⌵

Regola 2: modifichiamo il TTL e inoltriamo su tutte le altre porte.

Challenge: Provate a immaginare come dovrà essere compilato questo campo!

Mininet-GUI

Flow Control

FlowManager
SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

04:37:31 PM

http://localhost:8080/home/flowform.html

90%

Flow Control

1 switch available

+ Add Field

INSTRUCT

Goto Meter

APPLY ACTIONS

ACTION TYPE	VALUE
SET_FIELD	nw_ttl=100
OUTPUT	FLOOD

+ Add Action

☐ Clear Actions

e.g. OUTPUT

value

+ Add Action

Write Metadata

integer

Metadata Mask

integer

Goto Table

table_id

Regola 2: modifichiamo il TTL e inoltriamo su tutte le altre porte.

FlowManager
SDN APPLICATION

MONITOR

- Dashboard
- Flows
- Groups
- Meters
- Topology
- Messages

CONTROL

- + Flow Control**
- + Group Control
- + Meter Control

04:39:03 PM

Flow Control

1 switch available

Goto Meter

APPLY ACTIONS

ACTION TYPE	VALUE
SET_FIELD	nw_ttl=100
Set a "field" using "value" (The set of...	
OUTPUT	FLOOD
Output packet from "port"	
<button>+ Add Action</button>	

WRITE ACTIONS

ACTION TYPE	VALUE
e.g. OUTPUT	value
<button>+ Add Action</button>	
Write Metadata	integer
Metadata Mask	integer
Goto Table	table_id

Regola 2: Programmiamo la regola.

Flow Tables ...

Pictures - Th...

Terminal -

Mininet-GUI

Flow Tables

FlowManager
SDN APPLICATION

MONITOR

Dashboard

Flows

Groups

Meters

Topology

Messages

CONTROL

Flow Control

Group Control

Meter Control

SYSTEM

04:41:09 PM

Flow Tables

Updated 04:39:50 PM

Refresh

SW_1

Flow Table 3 entries

Actions

	PRIORITY	MATCH FIELDS							SET F	BYTE COUNT	FLAGS
☐	65535	eth_dst = 01:80:c2:00:00: eth_type = 35020								0	0
☐	100	eth_type = 2048	0	106	0	0	SET_FIELD: ipv4_ttl:100 OUTPUT:FLOOD	0	0	0	
☐	0	ANY	0	389	0	0	OUTPUT:FLOOD	4	280	0	

Con la funzione «Flows» è possibile vedere tutte le regole caricate.

Mininet-GUI

Flow Tables

Not Secure

http://10.0.2.15:4020

90%

Mininet GUI

File

View

Testing

Run Pingall Test

Run IPerf

Topology

Generate Topology

Connect Nodes

Delete Selected

Nodes Palette

h1

10.0.0.1/8

s1-eth1

s1-eth0

h2

10.0.0.2/8

h3

10.0.0.3/8

(Click-Destro) Apriamo una shell su H2...

Open WebShell

View Node Stats

Network Tools

WebShell

Sniffer

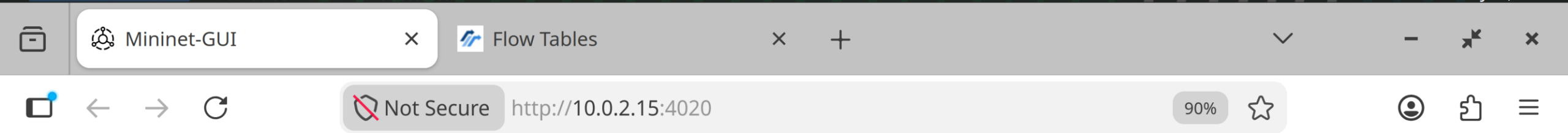
Monitoring

Logs

Connected to Mininet

Network Started

Hosts 2 · Controllers 1 · Switches 1



Mininet GUI File View

Testing

Run Pingall Test

Run IPerf

Topology

Generate Topology

Connect Nodes

Delete Selected

Nodes Palette

Network Tools

WebShell Sniffer Monitoring Logs

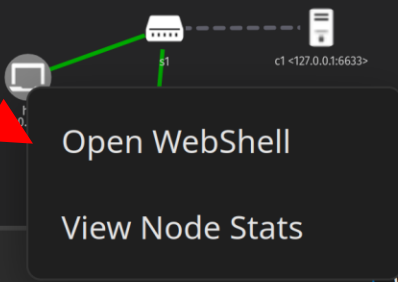
H2(2) x

```
root@h2:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ cd ..
root@h2:/home/sdn/Repos/mininet-gui$ cd utils/
root@h2:/home/sdn/Repos/mininet-gui/utils$ cd webserver-ttl/
root@h2:/home/sdn/Repos/mininet-gui/utils/webserver-ttl$ ./webserver-ttl
2026/06/08 17:38:17 listening on :2000 (TTL threshold: 100)
```

... e avviamo il server HTTP.

Si trova in ../utils/webserver-ttl/.
Avviato ascolta sulla porta TCP 2000.

(Click-Destro) Apriamo una shell su H1...



- Open WebShell
- View Node Stats

WebShell

Sniffer Monitoring Logs

Testing

- Run Pingall Test
- Run IPerf

Topology

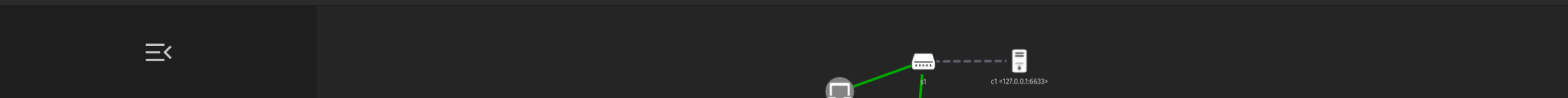
- Generate Topology
- Connect Nodes
- Delete Selected

Nodes Palette

Network Tools

H2(2) x

```
root@h2:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ cd ..
root@h2:/home/sdn/Repos/mininet-gui$ cd utils/
root@h2:/home/sdn/Repos/mininet-gui/utils$ cd webserver-ttl/
root@h2:/home/sdn/Repos/mininet-gui/utils/webserver-ttl$ ./webserver-ttl
2026/06/08 17:38:17 listening on :2000 (TTL threshold: 100)
```



Testing

Run Pingall Test

Run IPerf

WebShell Sniffer Monitoring Logs

H2(2) x H1(3) x

Topology

```
root@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ curl http://10.0.0.2:2000
```

Generate Topology

Connect Nodes

Delete Selected

Nodes Palette

Menu icon

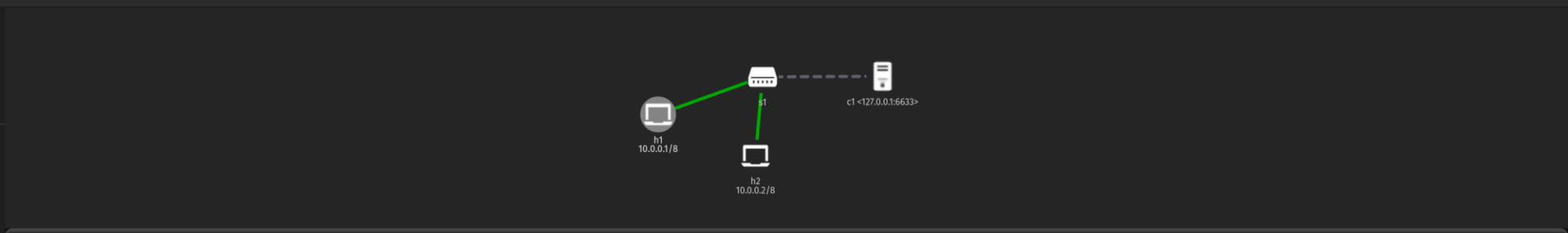
Testing

- Run Pingall Test
- Run IPerf

Topology

- Generate Topology
- Connect Nodes
- Delete Selected

Nodes Palette



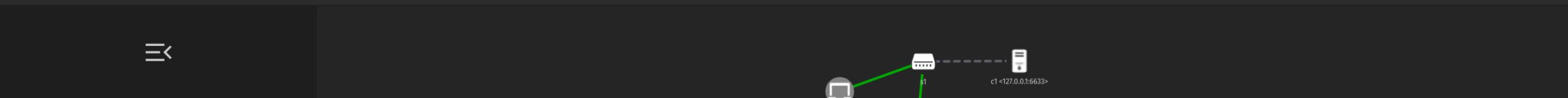
Network Tools

WebShell Sniffer Monitoring Logs

H2(2) x H1(3) x

```
root@h1: /home/sdn/Repos/mininet-gui/mininet-gui-backend$ curl http://10.0.0.2:2000
root@h1: /home/sdn/Repos/mininet-gui/mininet-gui-backend$
```





Testing

Run Pingall Test

Run IPerf

Network Tools WebShell Sniffer Monitoring Logs

H2(2) x H1(3) x

Topology

Generate Topology

Connect Nodes

Delete Selected

```
root@h1: /home/sdn/Repos/mininet-gui/mininet-gui-backend$ curl http://10.0.0.2:2000  
&>`0s7[0A~s`/TTLr root@h1: /home/sdn/Repos/mininet-gui/mininet-gui-backend$
```

Nodes Palette

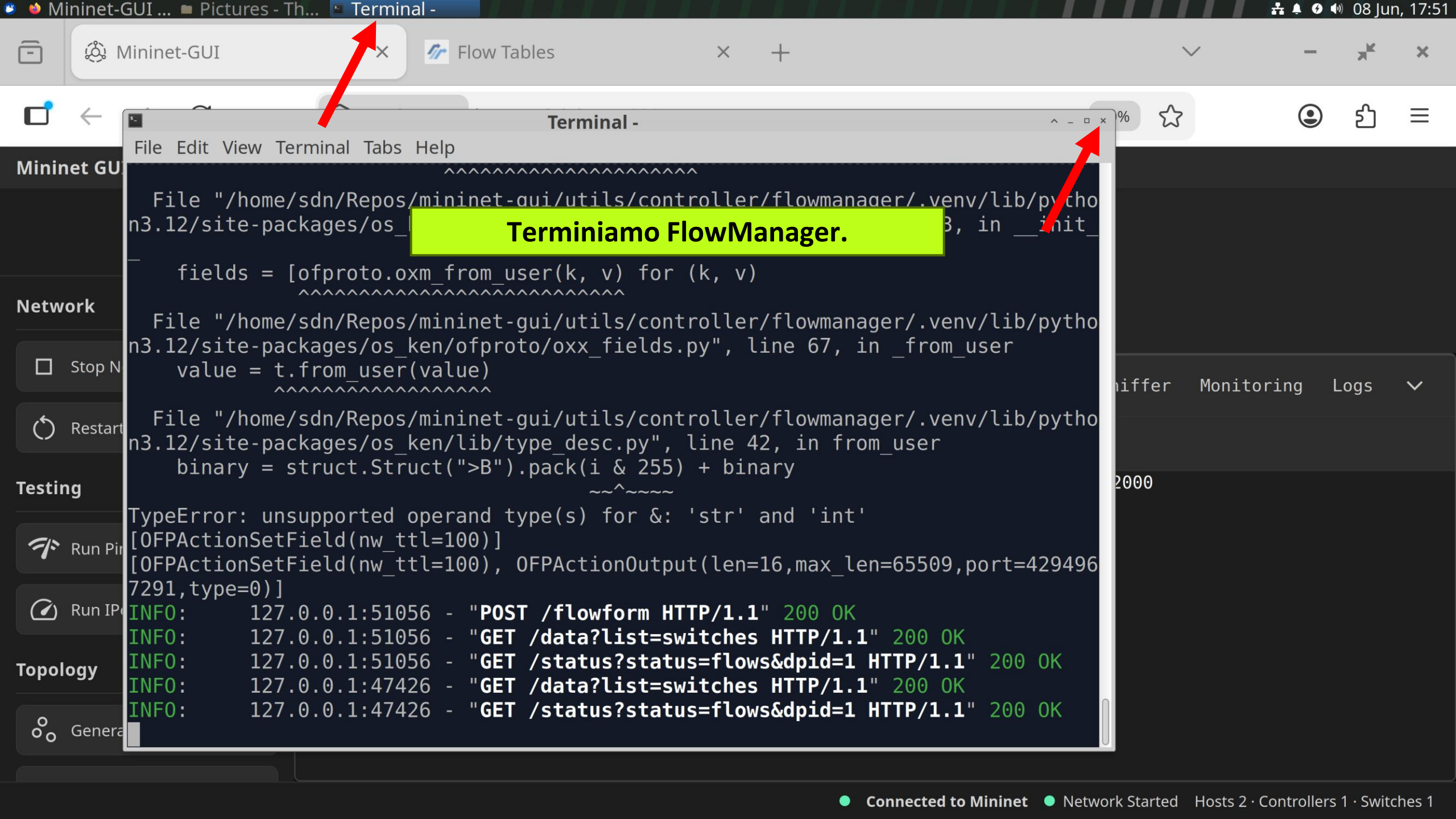


- Questa sarebbe SDN?
 - 💰 Automazione: Abbiamo letteralmente fatto tutto a mano.
 - 📉 Scalabilità: Abbiamo programmato uno switch come un hub... Al primo loop esplode tutta la rete...
- Cosa si salva?
 - Abbiamo fatto pilotare uno switch da un controller...
 - ...Ma un controller può pilotare tanti switch...
 - ...E il controller, oltre che manualmente, si può programmare (o farlo programmare 🤖) con delle API in un linguaggio di programmazione (eg. Python).
 - Programmare i dispositivi di rete diventa facile come programmare un qualunque software.
 - Uno switch può agire da L2 (Ethernet) fino a L4 (TCP/UDP) permettendo a questi dispositivi da agire da Router, Firewall, NAT Gateway, Load Balancer...



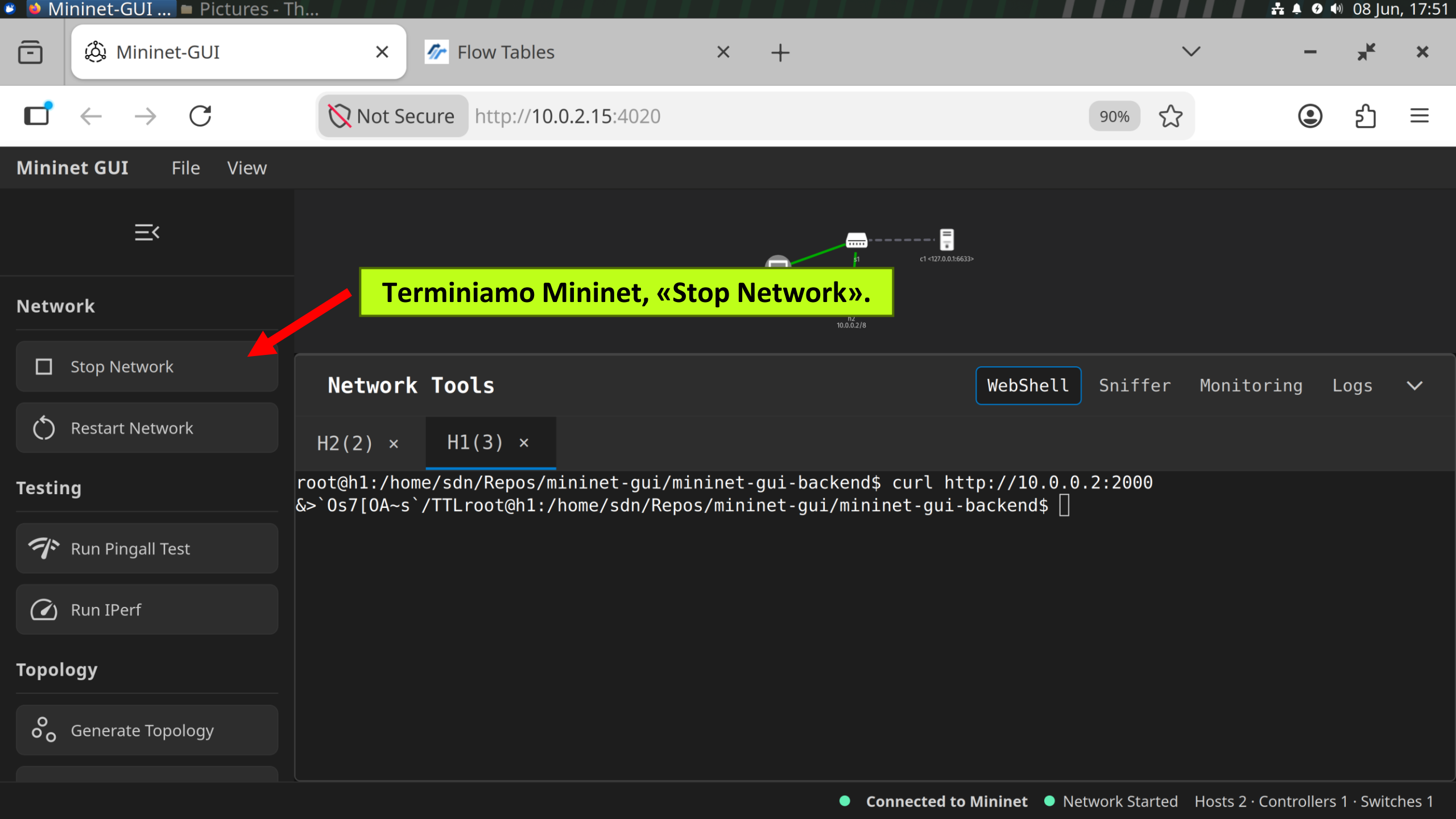
POLITECNICO
MILANO 1863

SDN Challenge #2



Terminiamo FlowManager.

```
File "/home/sdn/Repos/mininet-gui/utils/controller/flowmanager/.venv/lib/python3.12/site-packages/os_
fields = [ofproto.ofproto_from_user(k, v) for (k, v)
File "/home/sdn/Repos/mininet-gui/utils/controller/flowmanager/.venv/lib/python3.12/site-packages/os_ken/ofproto/ofproto.py", line 67, in _from_user
value = t.from_user(value)
File "/home/sdn/Repos/mininet-gui/utils/controller/flowmanager/.venv/lib/python3.12/site-packages/os_ken/lib/type_desc.py", line 42, in from_user
binary = struct.Struct(">B").pack(i & 255) + binary
TypeError: unsupported operand type(s) for &: 'str' and 'int'
[OFPActionSetField(nw_ttl=100)]
[OFPActionSetField(nw_ttl=100), OFPActionOutput(len=16,max_len=65509,port=4294967291,type=0)]
INFO: 127.0.0.1:51056 - "POST /flowform HTTP/1.1" 200 OK
INFO: 127.0.0.1:51056 - "GET /data?list=switches HTTP/1.1" 200 OK
INFO: 127.0.0.1:51056 - "GET /status?status=flows&dpid=1 HTTP/1.1" 200 OK
INFO: 127.0.0.1:47426 - "GET /data?list=switches HTTP/1.1" 200 OK
INFO: 127.0.0.1:47426 - "GET /status?status=flows&dpid=1 HTTP/1.1" 200 OK
```

Terminiamo Mininet, «Stop Network».

Network Tools

WebShell

Sniffer

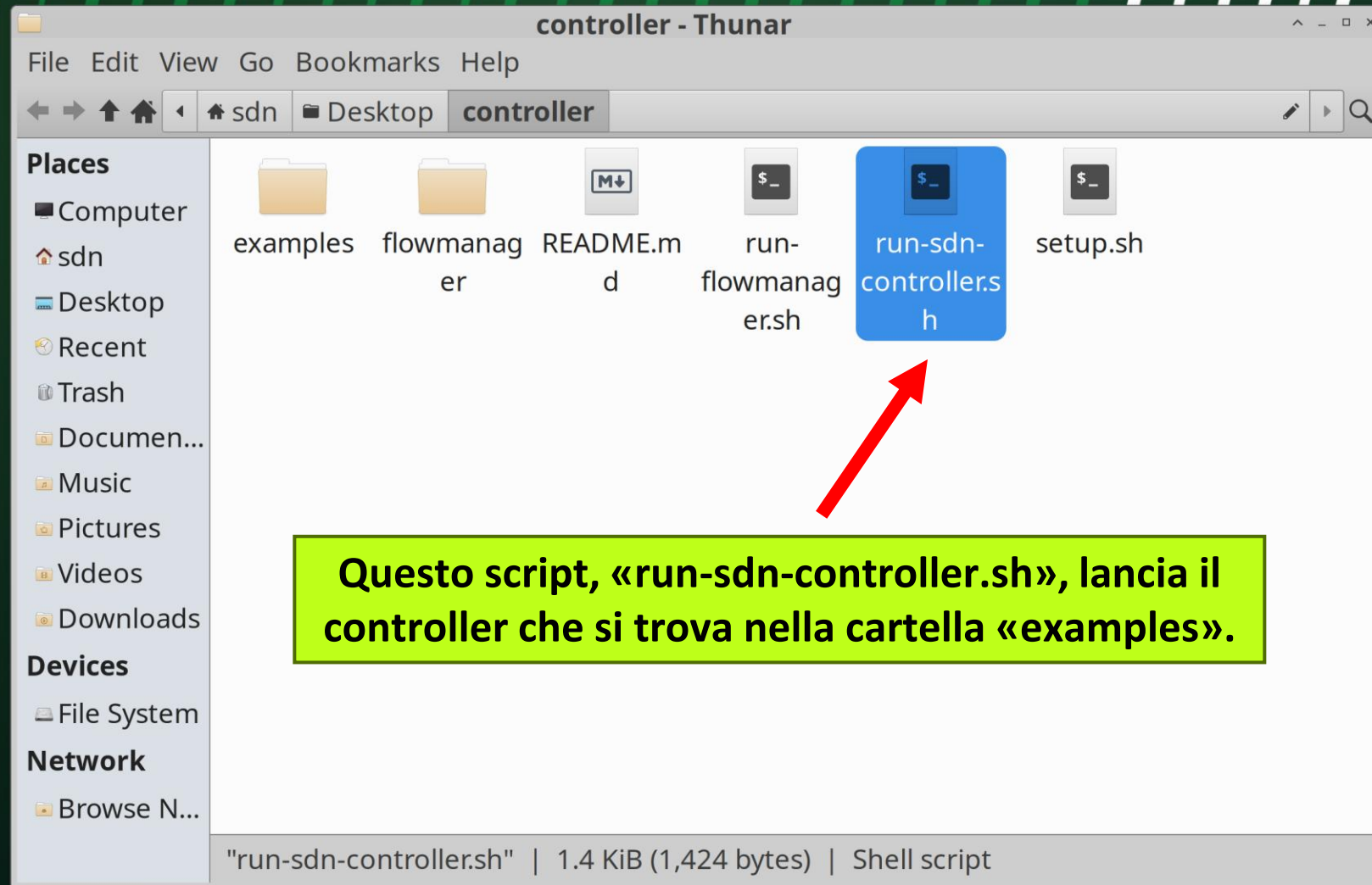
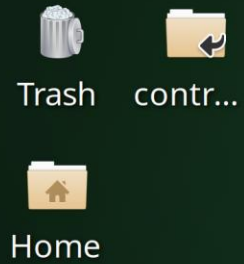
Monitoring

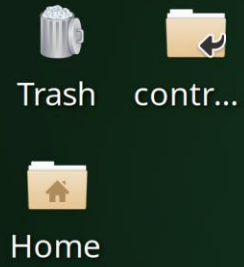
Logs

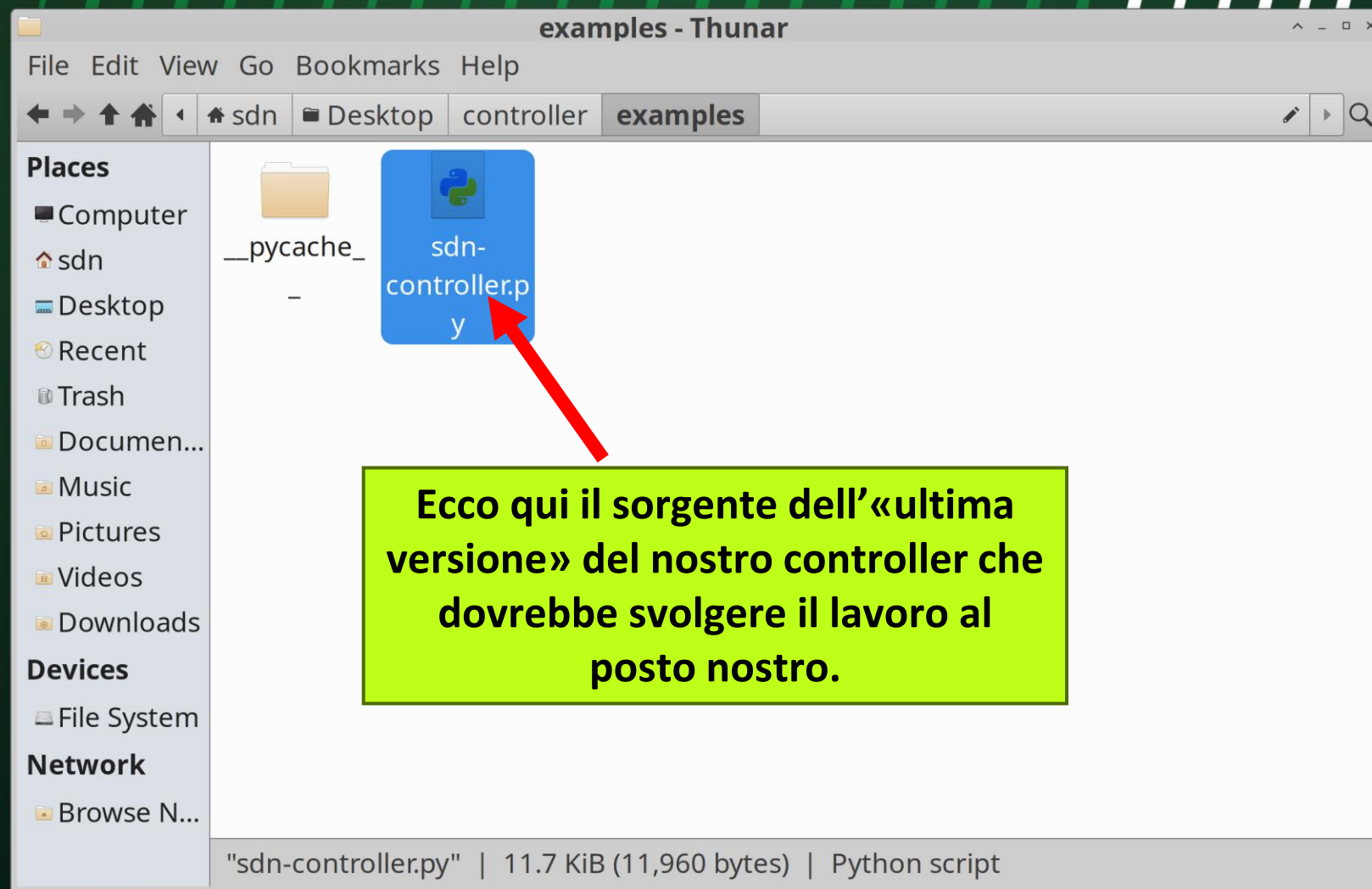
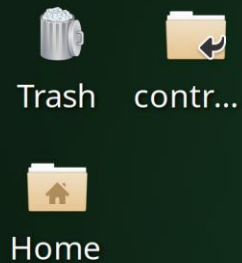
H2(2) x

H1(3) x

```
root@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ curl http://10.0.0.2:2000
&>`0s7[0A~s`/TTLroot@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend$
```







Ecco qui il sorgente dell'«ultima versione» del nostro controller che dovrebbe svolgere il lavoro al posto nostro.

File Edit Search View Document Help

```
# _actions – the single place where forwarding actions are built.
#
# Every time the controller tells a switch to send a packet somewhere,
# this method builds the list of OpenFlow actions.
#
# EXERCISE: modify this method so that every forwarded IPv4 packet
#           has its TTL set to 100.
# -----
```

```
def _actions(self, parser, output_port):
```

```
    """
```

```
    Build the list of OpenFlow actions.
```

```
    Called by both:
```

```
    - OFPPacketOut (forward the *current* packet – line 207)
```

```
    - OFPFlowMod    (install a rule for *future* packets – line 217)
```

```
    """
```

```
    return [
```

```
        ## parser.#####, You need to add something here.
```

```
        parser.OFPACTIONOutput(output_port),
```

```
    ]
```

```
# -----
```

```
# switch_features_handler
```

```
#
```

```
# Called once per switch, right after the OpenFlow handshake.
```

```
# Installs the *table-miss* flow: a catch-all rule with priority 0
```

```
# that sends every unmatched packet to the controller.
```

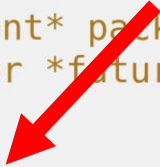
```
#
```

```
# OFPP_CONTROLLER = virtual port that means "send to the controller"
```

```
# OFPCML_NO_BUFFER = do not store the packet on the switch; send the
```

```
#
```

Challenge 2: Lo script è corrotto, manca una riga, come lo sistemiamo?




```
# _actions – the single place where forwarding actions are built.
#
# Every time the controller tells a switch to send a packet somewhere,
# this method builds the list of OpenFlow actions.
#
# EXERCISE: modify this method so that every forwarded IPv4 packet
#           has its TTL set to 100.
# -----
```

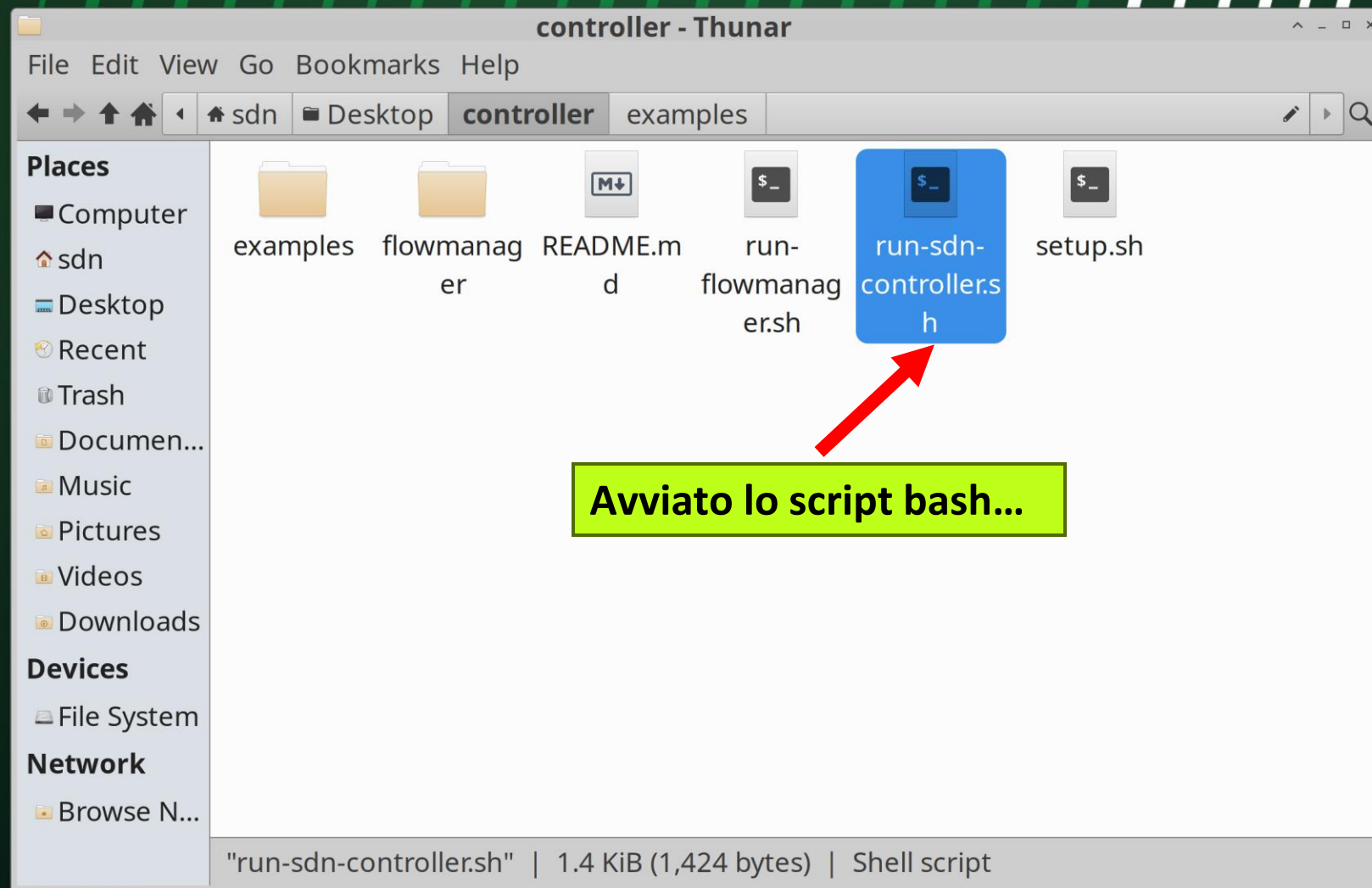
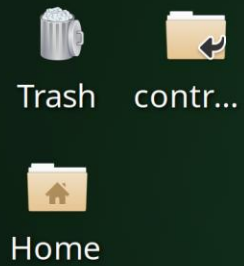
```
def _actions(self, parser, output_port):
```

Soluzione:

parser.OFPActionSetNwTtl(nw_ttl=100)

```
    # packet_out (forward the current packet – line 207)
    - OFPFlowMod (install a rule for *future* packets – line 217)
    """
    return [
        parser.OFPActionSetNwTtl(nw_ttl=100),
        parser.OFPActionOutput(output_port),
    ]
```

```
# -----
# switch_features_handler
#
# Called once per switch, right after the OpenFlow handshake.
# Installs the *table-miss* flow: a catch-all rule with priority 0
# that sends every unmatched packet to the controller.
#
# OFPP_CONTROLLER = virtual port that means "send to the controller"
# OFPCML_NO_BUFFER = do not store the packet on the switch; send the
```



Mininet-GUI

FlowManager

Not Secure

http://10.0.2.15:4020

90%

Mininet GUI

File

View

Network

Start Network

Restart Network

Testing

Run Pingall Test

Run IPerf

Topology

Generate Topology

h1
10.0.0.1/8

1

c1 <127.0.0.1:6633>

...possiamo avviare nuovamente Mininet.

Network Tools

WebShell

Sniffer

Monitoring

Logs

2026-06-08 15:43:35,994 Level 25 h1

2026-06-08 15:43:35,994 Level 25

2026-06-08 15:43:35,994 Level 25 *** Results:

2026-06-08 15:43:35,995 Level 25 h1->h2: 1/1,

2026-06-08 15:43:35,995 Level 25 rtt min/avg/max/mdev 39.436/39.436/39.436/0.000 ms

2026-06-08 15:43:35,995 Level 25 h2->h1: 1/1,

2026-06-08 15:43:35,995 Level 25 rtt min/avg/max/mdev 20.388/20.388/20.388/0.000 ms

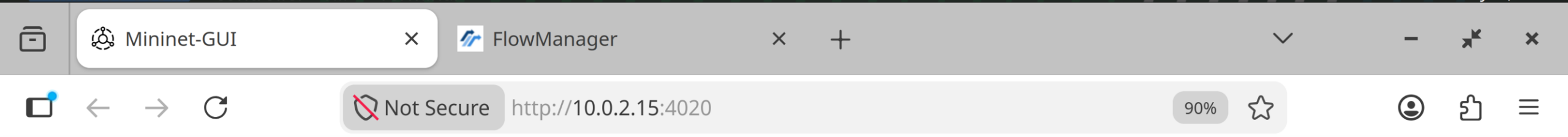
2026-06-08 16:28:07,539 WARNING Unable to contact the remote controller at 127.0.0.1:6633

2026-06-08 17:52:08,517 WARNING Unable to contact the remote controller at 127.0.0.1:6633

Connected to Mininet

Network Stopped

Hosts 2 · Controllers 1 · Switches 1



Mininet GUI File View



Network

☐ Stop Network

Restart Network

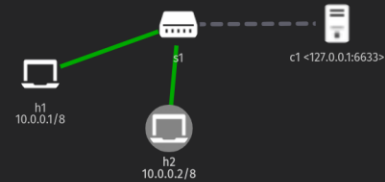
Testing

Run Pingall Test

Run IPerf

Topology

Generate Topology



Network Tools

Avviamo, di nuovo, il webserver su H2.

Sniffer

Monitoring

Logs



H2(3) x

```
root@h2:/home/sdn/Repos/mininet-gui/mininet-gui-backend$ cd ..
root@h2:/home/sdn/Repos/mininet-gui$ cd utils/
root@h2:/home/sdn/Repos/mininet-gui/utils$ cd webserver-ttl/
root@h2:/home/sdn/Repos/mininet-gui/utils/webserver-ttl$ ./webserver-ttl
2026/06/08 18:19:57 listening on :2000 (TTL threshold: 100)
```

Mininet-GUI

FlowManager

Not Securehttp://10.0.2.15:402090%

Mininet GUIFileView

Network

Stop Network

Restart Network

Testing

Run Pingall Test

Run IPerf

Topology

h1
10.0.0.1/8

s1

c1 <127.0.0.1:6633>

Proviamo a fare una richiesta HTTP da H1 a H2.
Dovremmo ottenere il nostro token.

H2(3) ×H1(4) ×

root@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend\$ curl http://10.0.0.2:2000
dDg2_R93xs.~,TTLroot@h1:/home/sdn/Repos/mininet-gui/mininet-gui-backend\$

Shell

Sniffer

Monitoring

Logs

N.B: In caso di errori di rete (eg. time-out) provate a far fare alle macchine («PingAll») traffico in modo che il controller possa scoprire la loro esistenza.

Connected to Mininet

Network Started

Hosts 2 · Controllers 1 · Switches 1